

Engine instruments in a split-shaft/free turbine engine typically consist of the following basic indicators.
 [Figure 15-7]

1. ITT indicator
2. Torquemeter
3. Propeller tachometer
4. N_1 (gas generator) tachometer
5. Fuel flow indicator
6. Oil temperature/pressure indicator



Figure 15-7. Engine instruments—split shaft/free turbine engine.

The ITT indicator gives an instantaneous reading of engine gas temperature between the compressor turbine and the power turbines. The torquemeter responds to power lever movement and gives an indication in foot-pounds (ft/lb) of the torque being applied to the propeller. Because in the free turbine engine the propeller is not attached physically to the shaft of the gas turbine engine, two tachometers are justified—one for the propeller and one for the gas generator. The propeller tachometer is read directly in revolutions per minute. The N_1 or gas generator is read in percent of rpm. In the Pratt & Whitney PT-6 engine, it is based on a figure of 37,000 rpm at 100 percent. Maximum continuous gas generator is limited to 38,100 rpm or 101.5 percent N_1 .

The ITT indicator and torquemeter are used to set takeoff power. Climb and cruise power are established with the torquemeter and propeller tachometer while observing ITT limits. Gas generator (N_1) operation is monitored by the gas generator tachometer. Proper observation and interpretation of these instruments provide an indication of engine performance and condition.

Reverse Thrust and Beta Range Operations

The thrust that a propeller provides is a function of the angle of attack (AOA) at which the air strikes the blades, and the speed at which this occurs. The AOA varies with the pitch angle of the propeller.

Forward pitch produces forward thrust—higher pitch angles being required at higher airplane speeds. [Figure 15-8A] So called “flat pitch,” shown in Figure 15-8B, is the blade position offering minimum resistance to rotation and no net thrust for moving the airplane.

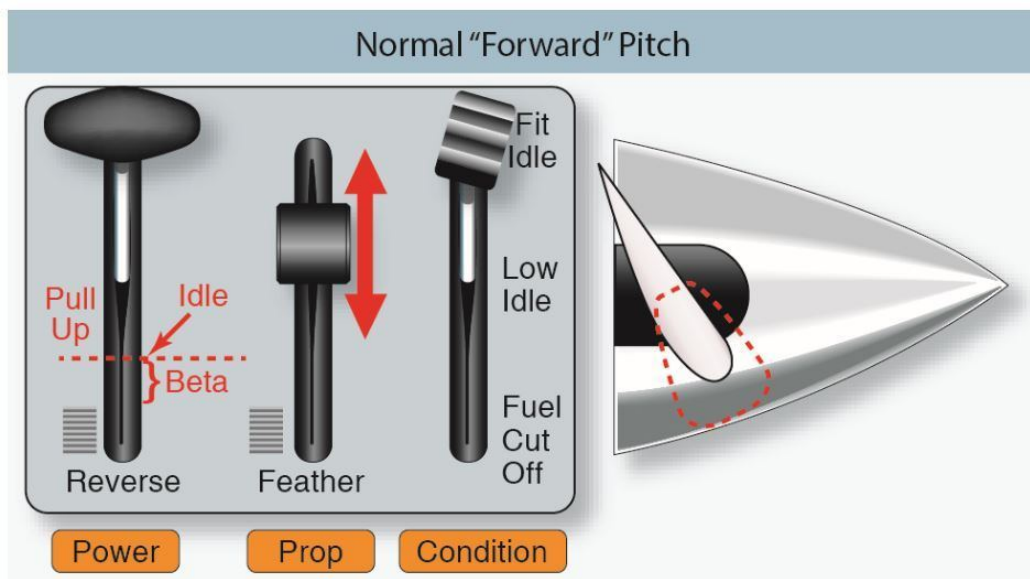


Figure 15-8A. Propeller forward pitch angle characteristics.

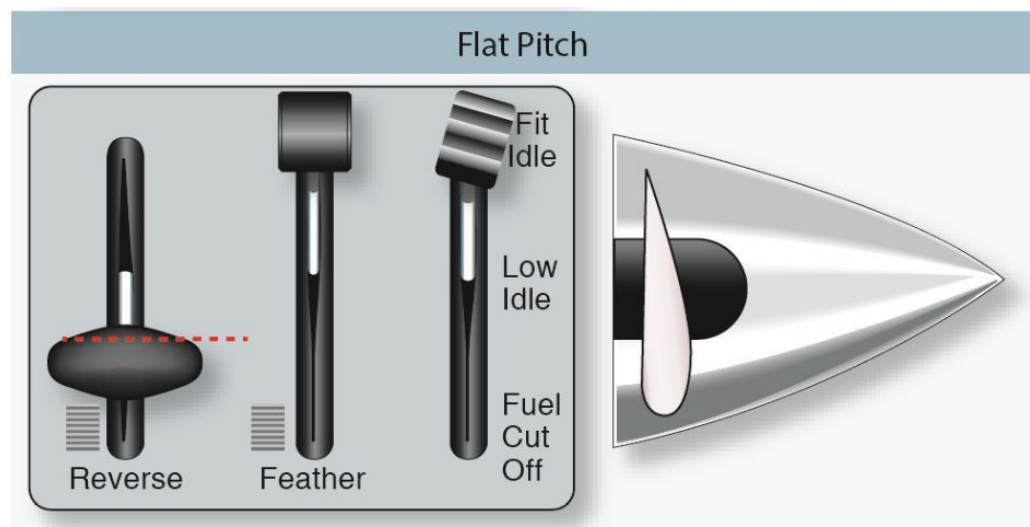


Figure 15-8B. Propeller flat pitch characteristics.

The “feathered” position is the highest pitch angle obtainable. [Figure 15-8C] The feathered position produces no forward thrust. The propeller is generally placed in feather only in case of in-flight engine failure to minimize drag and prevent the air from using the propeller as a turbine.

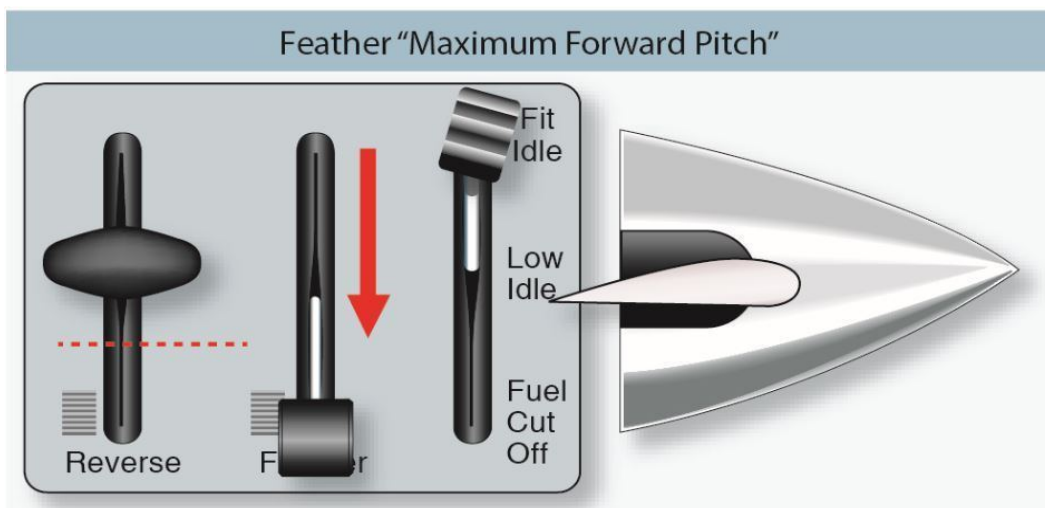


Figure 15-8C. Propeller feather (maximum forward pitch angle) characteristics.

In the “reverse” pitch position, the engine/propeller turns in the same direction as in the normal (forward) pitch position, but the propeller blade angle is positioned to the other side of flat pitch. [Figure 15-8D] In reverse pitch, air is pushed away from the airplane rather than being drawn over it. Reverse pitch results in braking action, rather than forward thrust of the airplane. It is used for backing away from obstacles when taxiing, controlling taxi speed, or to aid in bringing the airplane to a stop during the landing roll. Reverse pitch does not mean reverse rotation of the engine. The engine delivers power just the same, no matter which side of flat pitch the propeller blades are positioned.

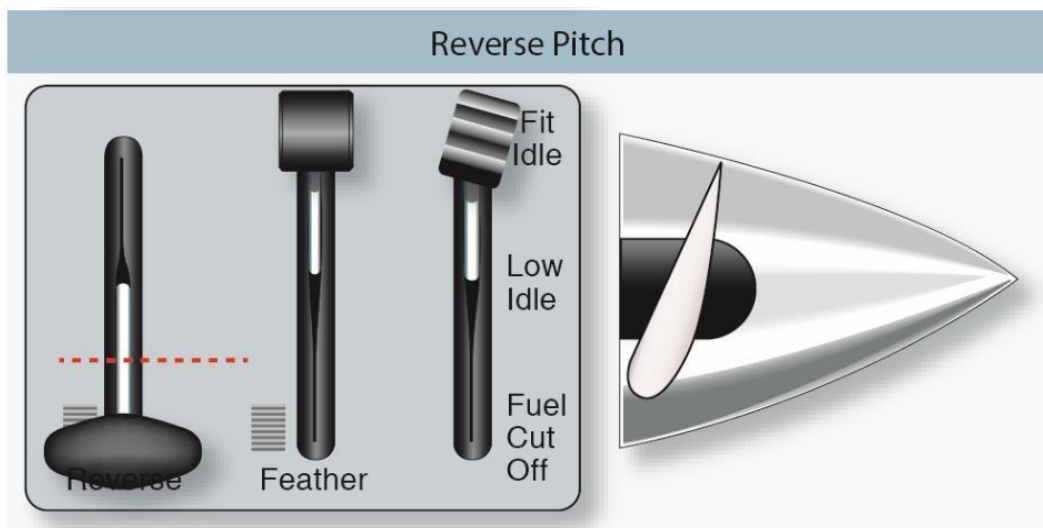


Figure 15-8D. Propeller reverse pitch characteristics.

With a turboprop engine, in order to obtain enough power for flight, the power lever is placed somewhere between flight idle (in some engines referred to as “high idle”) and maximum. The power lever directs signals to a fuel control unit to manually select fuel. The propeller governor selects the propeller pitch needed to keep the propeller/engine on speed. This is referred to as the propeller governing or “alpha” mode of operation. When positioned aft of flight idle, however, the power lever directly controls propeller blade angle. This is known as the “beta” range of operation.

The beta range of operation consists of power lever positions from flight idle to maximum reverse. Beginning at power lever positions just aft of flight idle, propeller blade pitch angles become progressively flatter with aft movement of the power lever until they go beyond maximum flat pitch and into negative pitch, resulting in reverse thrust. While in a fixed-shaft/constant-speed engine, the engine speed remains largely unchanged as the propeller blade angles achieve their negative values. On the split-shaft PT-6 engine, as the negative 5° position is reached, further aft movement of the power lever also results in a progressive increase in engine (N_1) rpm until a maximum value of about negative 11° of blade angle and 85 percent N_1 are achieved.

Operating in the beta range and/or with reverse thrust requires specific techniques and procedures depending on the particular airplane make and model. Specific engine parameters and limitations for operations within this area should be adhered to. It is essential that a pilot transitioning to turboprop airplanes becomes knowledgeable and proficient in these areas, which are unique to turbine-engine powered airplanes.

Turboprop Airplane Electrical Systems

The typical turboprop airplane electrical system is a 28-volt direct current (DC) system, which receives power from one or more batteries and a starter/generator for each engine. The batteries are either lead-acid, nickel-cadmium (NiCad), or Lithium-ion. When battery voltage is low, its ability to turn the compressor for engine start is greatly diminished, and the possibility of engine damage due to a hot start increases. Therefore, it is essential to check the battery's condition before every engine start. The different battery types have different operating characteristics depending on the specific aircraft installation and operational environment.

The DC generators used in turboprop airplanes double as starter motors and are called “starter/generators.” The starter/generator uses electrical power to produce mechanical torque to start the engine and then uses the engine's mechanical torque to produce electrical power after the engine is running. Some of the DC power produced is changed to 28 volt 400 cycle alternating current (AC) power for certain avionic, lighting, and indicator synchronization functions. This is accomplished by an electrical component called an inverter.

The distribution of DC and AC power throughout the system is accomplished through the use of power distribution buses. These “buses” as they are called are actually common terminals from which individual electrical circuits get their power. [Figure 15-9]

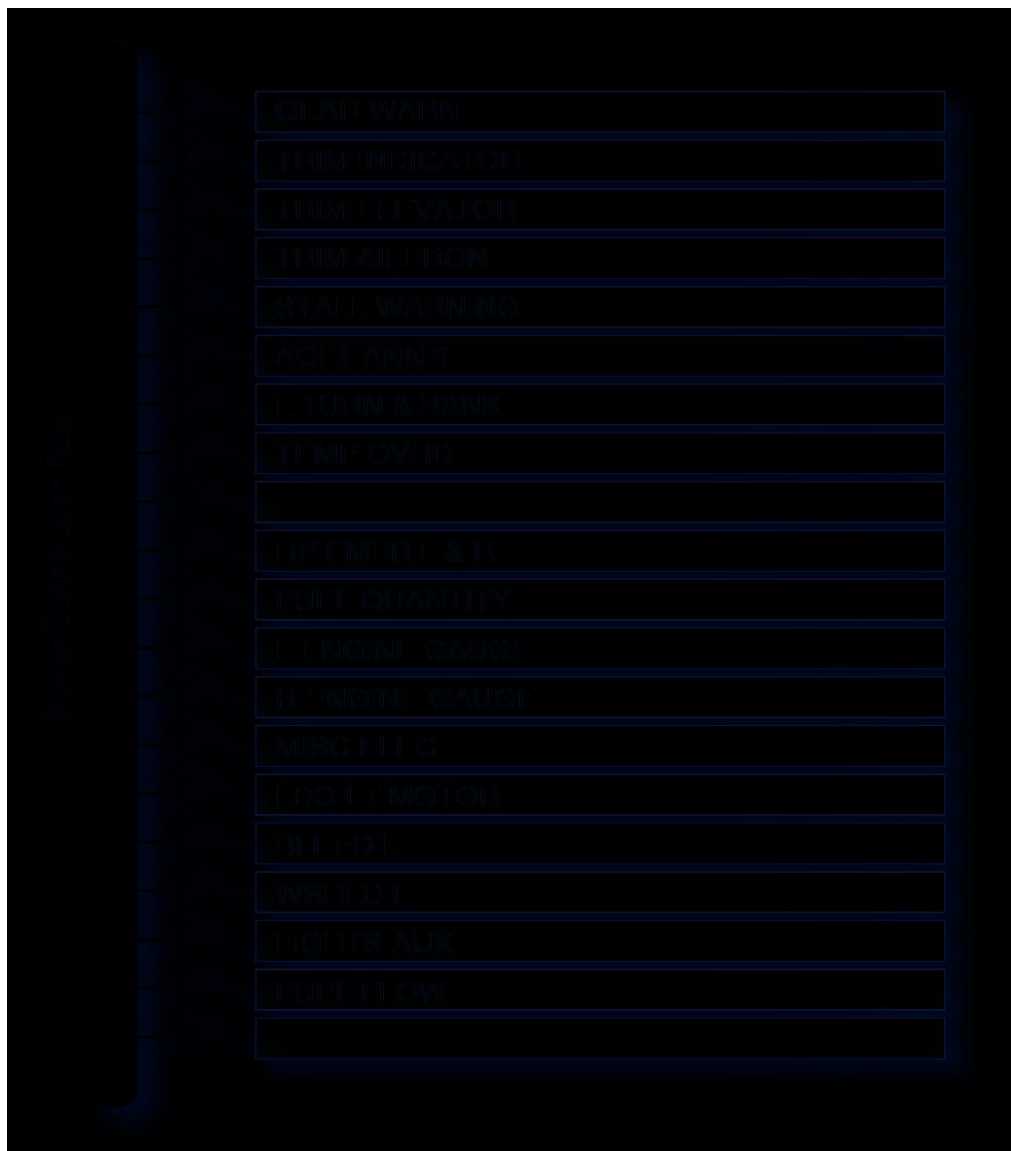


Figure 15-9. Typical individual power distribution bus.

Busess are usually named for what they power (avionics bus, for example) or for where they get their power (right generator bus, battery bus). The distribution of DC and AC power is often divided into functional groups (buses) that give priority to certain equipment during normal and emergency operations. Main buses serve most of the airplane's electrical equipment. Essential buses feed power to equipment having top priority. [Figure 15-10]

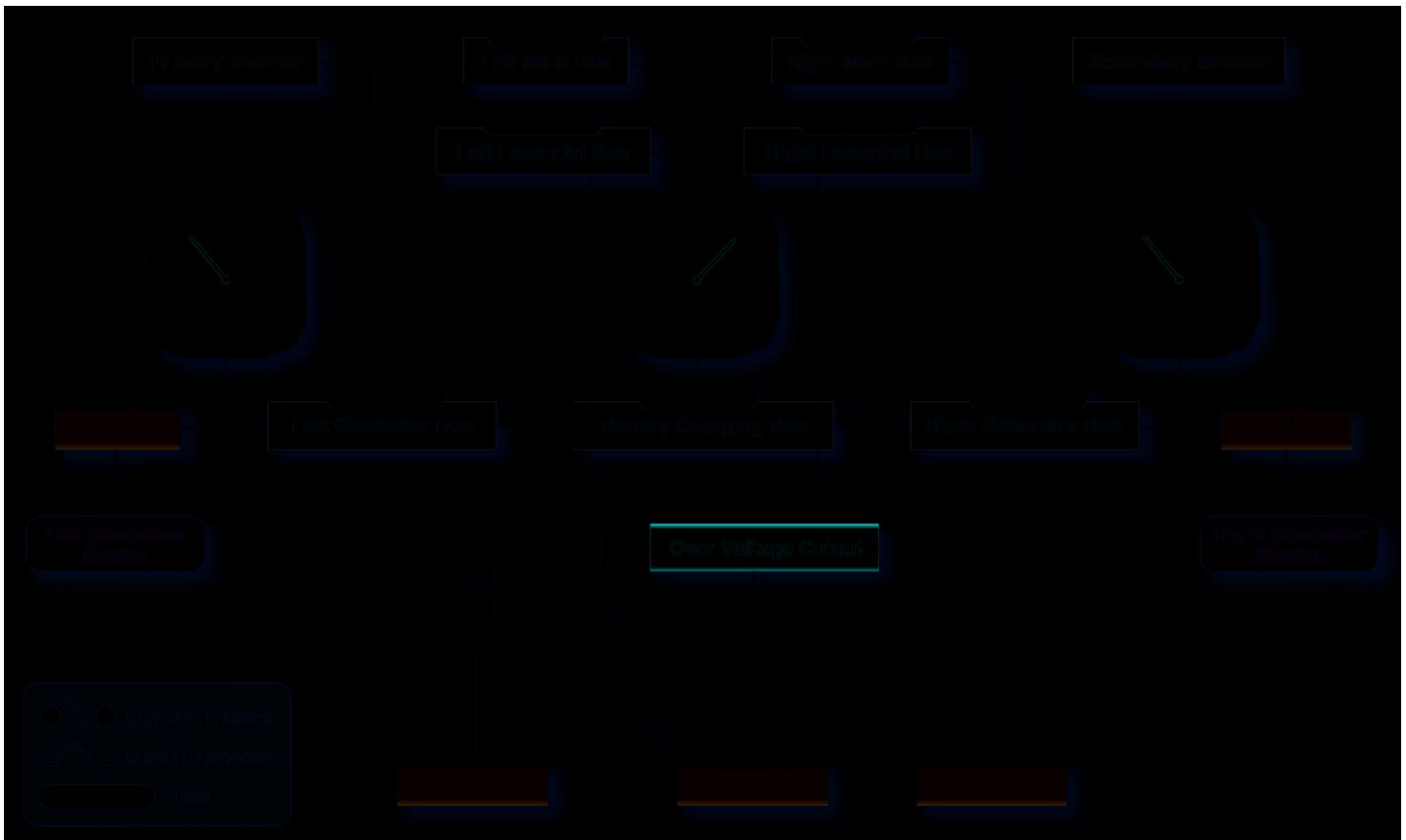


Figure 15-10. Simplified schematic of turboprop airplane electrical system.

Multiengine turboprop airplanes normally have several power sources—at least one generator per engine and at least one battery for the airplane. The electrical systems are usually designed so that any bus can be energized by any of the power sources. For example, a typical system has a left and right engine generator-powered bus. While these buses are normally isolated, they may be fed from other power sources. However, in the event of a short-circuit, the bus remains isolated. Pilots should refer to the appropriate checklist when an electrical fault occurs.

Power distribution buses are protected from short circuits and other malfunctions by a type of fuse called a current limiter. In the case of excessive current supplied by any power source, the current limiter opens the circuit and thereby isolates that power source and separates the affected bus from the system. If this occurs, pilots should refer to the appropriate checklist.

Operational Considerations

As previously stated, a turboprop airplane flies just like any other piston engine airplane of comparable size and weight. It is the operation of the engines and airplane systems that makes the turboprop airplane different from its piston engine counterpart. Pilot errors in engine and/or systems operation are common causes of aircraft damage or loss of aircraft control. There are two engine-related issues that should be considered when a pilot transitions to turboprop operations.

The first issue concerns the split-shaft/free turbine engine, where power output lags for several seconds when the pilot moves the power lever from flight idle to a high power setting. This delay may surprise a pilot who has only flown airplanes with a piston engine (or a fixed-shaft turboprop). Certain operations such as firefighting and agricultural application require maneuvering close to the ground while operating at or near flight idle. Although smooth power applications are still the rule, the pilot should be aware that a greater physical movement of the power levers is required as compared to throttle movement in a piston engine. The pilot should understand the lag and anticipate and lead the power changes more than in the past and should keep in mind that the last 30 percent of engine rpm represents the majority of the engine thrust. Below that setting, the application of power has very little effect.